

Q.1 what are weak learners?

→ weak learners are simple models in ML that perform just better than random chance. They are usually high bias, low complexity, models meaning they don't capture all patterns in data and individually make many errors.

Q.2 How is random forest related to decision tree?

→ Random forest is directly related to decision tree

i) decision tree: Single tree that makes prediction by splitting data as liked on feedback

ii) Random forest: An ensemble of decision trees. Each tree is trained on random forest subset of data and a random subset of features.

A random forest is like a team of decision trees working together.

Q.3 What is difference between bagging and boosting?

Feature	Bagging	Boosting
i) Goal	Reduce variance	Reduce bias, variance
ii) Method	Train model in parallel	Train model in sequential
iii) Example	Random forest	AdaBoost, GB
iv) Voting	Majoring vote	weight vote